EEG-Twin: An Interpretable, Trajectory-Aware Digital-Twin Framework for Neurological Outcome Prediction After Cardiac Arrest

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MAIN FINDING

From the first hours of EEG, a causal digital twin predicts who will recover, simulates what comes next, and shows the real patients it reasons from.

0.94
AUROC, full recording
95% CI 0.91–0.96 · n = 299 · Fig. 2A

0.91
AUROC with only 12 h of EEG
n = 98 with EEG by 12 h · Fig. 2A

78%
of poor outcomes identified
at a false-positive rate ≤ 5%

0.94
AUROC from 10 similar patients
vote of known outcomes · Fig. 5

INTRODUCTION

- After cardiac arrest, prognosis is often uncertain, and ambiguous signs can push care toward early withdrawal.
- EEG evolves over days, but most models give one static score.
- **Objective:** a digital twin that updates hourly, forecasts the EEG, and grounds each prediction in similar prior patients.

METHODS

- **Data:** I-CARE (PhysioNet 2023): 7 hospitals, 994 patients, 525,705 five-minute EEG segments.
- **Outcome:** CPC 1–2 (good) vs 3–5 (poor) at 3–6 months.
- **Evaluation:** 695 development patients (254 good); 299 held-out test patients (102 good). 95% CIs by patient bootstrap.

1 Segment Continuous EEG is cut into 5-minute segments over the first 84 h after ROSC.

2 Phenotype ProtoPNet scores each segment against 45 learned EEG prototypes and assigns 1 of 8 ACNS states.

3 Embed CEBRA maps each segment to a point on a 3D sphere, so a recording becomes a visible path (Fig. 3).

4 Summarize Each hour is reduced to 98 interpretable features, plus 5 admission variables.

5 Predict + forecast A causal Transformer (10-model ensemble) gives P(good outcome) every hour and forecasts the next hour of EEG (Fig. 2, 4).

6 Retrieve The 10 most similar prior patients, as of that hour, are returned with their known outcomes (Fig. 5).

Figure 1. The EEG-Twin pipeline. Steps 1–6 are repeated every hour for each patient, using only EEG recorded up to that hour.

RESULTS

EEG-Twin is accurate at any hour and matches a model that sees the whole recording

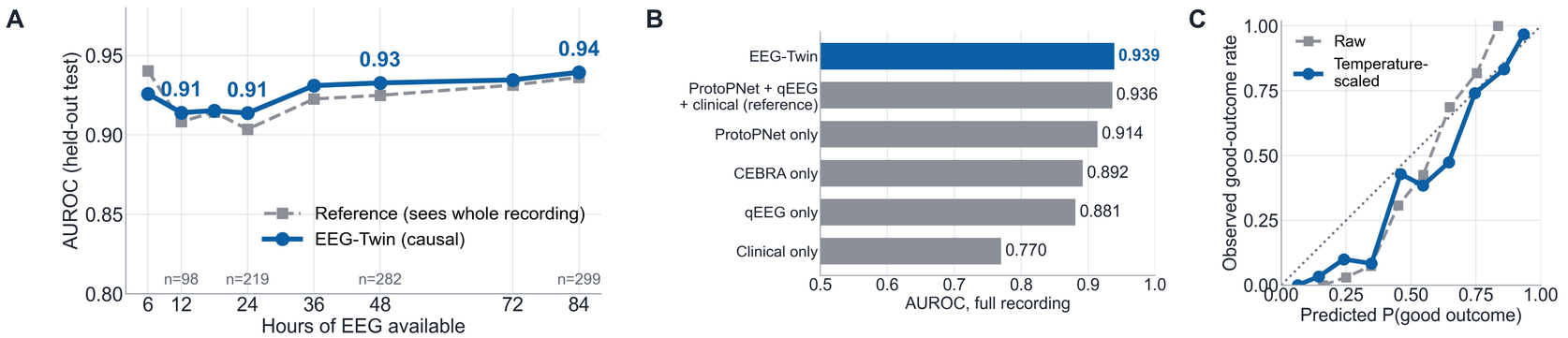


Figure 2. Outcome prediction on the held-out test set (n = 299). **(A)** AUROC by hours of EEG available; twin minus reference at 84 h: +0.003 (95% CI −0.008 to 0.015). **(B)** Full-recording AUROC by model input; clinical variables alone reach 0.77. **(C)** Calibration; Brier score 0.122 raw, 0.104 after temperature scaling.

Recovery and non-recovery trace different paths through EEG-state space

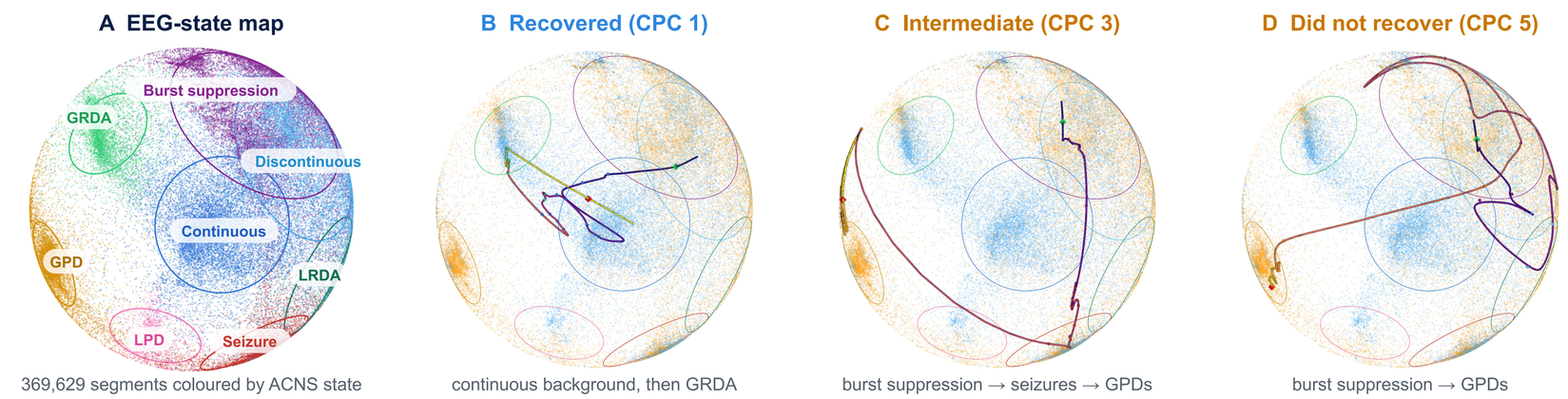


Figure 3. CEBRA trajectory space. **(A)** Every training segment, coloured by EEG state. **(B–D)** One held-out patient per outcome grade; the text under each globe gives the patient's dominant EEG states in order.

It simulates the future from early EEG

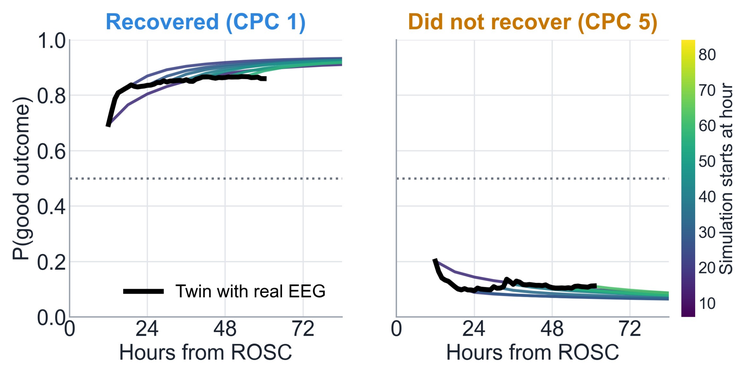


Figure 4. Roll-forward simulation in two held-out patients. Each coloured line **stops using real EEG** at its start hour and feeds on its own forecasts. Across the test set, AUROC simulated forward from 12 h is 0.916 (0.914 read directly); next-hour forecasts beat "no change" by 4.3% for EEG-state frequencies, 1.1% overall.

It finds real patients like this one

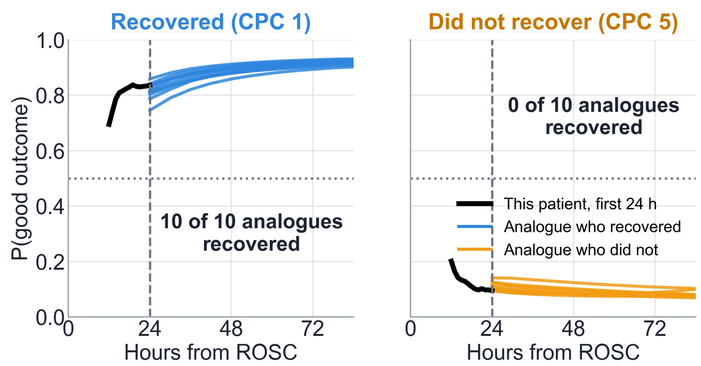


Figure 5. The 10 most similar prior patients at hour 24, for the same two patients as Fig. 4. Across the test set, **when 0–1 of a patient's 10 analogues recovered, 4% recovered (n = 104); when 8–10 did, 85% did (n = 33)** (hour 24, n = 219).

CONCLUSIONS

- **One causal model** gives an hourly outcome probability, an EEG forecast, and matched prior patients.
- **It matches a non-causal reference** (AUROC 0.94) and already reaches 0.91 with 12 h of EEG.
- **Proof of concept** on one retrospective split. Next: external and prospective bedside validation.

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